

Ioanna Vavelidou

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EDUCATION	PhD in Electrical Engineering Stanford University, CA Advisor: Caroline Trippel, High Assurance Computer Architectures Lab	<i>Sept 2021 - Present</i>
	MS in Electrical Engineering Stanford University, CA	<i>Sept 2022 - June 2024</i>
	Diploma in Electrical and Computer Engineering (Joint BSc & MEng) National Technical University of Athens, Greece Advisor: Professor Nectarios Koziris, Computing Systems Laboratory Thesis: <i>Efficient Shared Resource Management in Multiprocessors using Reinforcement Learning techniques</i> GPA: 9.26/10 (top 3%) Department Entrance Rank: 1st (Highest score in the National University Admission Exams, top 0.01% nation-wide)	<i>Oct 2015 - July 2021</i>
RESEARCH EXPERIENCE	Ph.D Research Assistant Stanford University, CA Advised by Professor Caroline Trippel Topic: <i>Reliable Datacenter Infrastructure Through Defect Detection and Mitigation</i>	<i>April 2022 - Present</i>
	Undergraduate Researcher National Technical University of Athens, Greece Thesis: <i>Efficient Shared Resource Management in Multiprocessors using Reinforcement Learning techniques</i>	<i>Dec 2019 - July 2021</i>
INTERNSHIPS	Student Researcher at Google Platforms <i>Developing an ML-driven probability model for predicting insidious hardware failures in Google's AI chip fleet, enabling smarter testing and performance trade-off decisions.</i>	<i>Feb 2026 - December 2026</i>
	Student Researcher at Google DeepMind <i>Deployed and evaluated real-world ML workloads as hardware reliability tests, identifying which workloads best expose defect-induced errors in AI accelerators.</i>	<i>June 2025 - Dec 2025</i>
	Student Researcher at Google Platforms <i>Designed compiler-based functional tests for Silent Data Corruption detection in CPUs.</i>	<i>October 2023 - June 2024</i>
	PhD Software Engineer Intern at Meta Platforms <i>Optimized memory usage for ML workloads to improve performance and scalability</i>	<i>June 2022 - Sept 2022</i>
HONORS & AWARDS	Award by the President of the Republic of Greece <i>For achieving the highest National University Admission score in my educational district (Attica district A')</i>	
	Award and Grant by the National Technical University of Athens <i>For achieving the highest entrance score in my department</i>	
	Award and Grant by Eurobank EFG, "The Great Moment for Education" <i>For achieving the highest National University Admission score in my High School</i>	

PUBLICATIONS	Ioanna Vavelidou, Subho Banerjee, Eric X. Liu, Mike Fuller, Subhasish Mitra, Caroline Trippel ITHICA: Intra-Thread Instruction Checking Approach for Defect-Induced Data Corruptions <i>MICRO 2026, IEEE/ACM International Symposium on Microarchitecture</i> Saranyu Chattopadhyay, Keerthikumara Devarajegowda, Bihan Zhao, Florian Lonsing, Brandon Alan D’Agostino, Ioanna Vavelidou, Vijay Deep Bhatt, Sebastian Prebeck, Wolfgang Ecker, Caroline Trippel, Clark Barrett, Subhasish Mitra G-QED: Generalized QED Pre-silicon Verification beyond Non-Interfering Hardware Accelerators <i>DAC 2023, Design Automation Conference</i> Hang Qiu, Ioanna Vavelidou, Jian Li, Evgenya Pergament, Pete Warden, Sandeep Chinchali, Zain Asgar, Sachin Katti ML-EXRAY: Visibility into ML Deployment on the Edge <i>MLSys 2022, Conference on Machine Learning and Systems</i>
MENTORSHIP	CURIS (Summer 2025): Designed and mentored a Stanford CS undergraduate research project on hardware fault injection in ML models. SURA Mentorship 2026: Mentoring two early-stage undergraduate researchers through Stanford Undergraduate Research Association (SURA). WEE Mentorship (2025-now): Mentoring an undergraduate woman in EE through Stanford’s Women in Electrical Engineering mentorship program. Stanford Science Penpals (2024-now): Exchanging monthly letters with a K-12 female student to share the experience of being a STEM scientist.
SKILLS	Languages: English (highly proficient, Michigan ECPE C2 Degree, TOEFL), French (adequate working knowledge, Delf B2 Degree), Greek (native) Programming Languages: Python, C, C++, MATLAB, x86 Assembly, LLVM IR Relevant Coursework: Machine Learning on Embedded Systems, Reinforcement Learning, Human-Computer Interaction Research, Cloud Computing Research, Principles of Data-Intensive Systems, Performance Engineering of Computer Systems & Networks, Computer Systems Architecture (Stanford), Speech and Natural Language Processing, Pattern Recognition, Computer Vision, Advanced Robotics, Advanced Topics in Computer Architecture, Logic Design of Digital Systems, Operating Systems, Networks and Circuitry Theory, VLSI System Design (NTUA)
GRE SCORES	GRE General Aug. 2020: Quantitative Reasoning 169/170, Verbal Reasoning 163/170, Analytical Writing 4.5/6
PROFESSIONAL MEMBERSHIPS	• ACM Member, ACM SIGARCH • Computing Research Association - Widening Participation (CRA-WP) • IEEE Member